

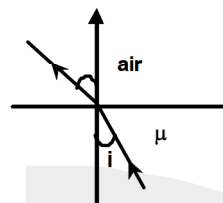
HINTS AND SOLUTIONS

Q.1 D

Sol. $\frac{\sin i}{\sin r} = \frac{1}{\mu} \Rightarrow \sin i = \frac{\sin r}{\mu}$

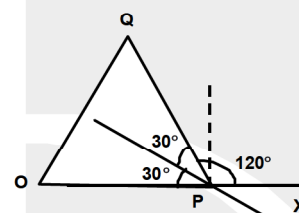
$$\Rightarrow \sin i = \frac{2 \sin i \cos i}{\mu}$$

$$\Rightarrow i = \cos^{-1}(\mu/2) \Rightarrow r = 2 \cos^{-1}(\mu/2)$$



Q.2 C

Sol. x-coordinate of the point Q is
 $PQ \cos 120^\circ = 20 \cos 120^\circ = -10 \text{ cm}$
 y-coordinate of the point Q is
 $PQ \sin 120^\circ = 20 \sin 120^\circ = 10\sqrt{3} \text{ cm}$



Q.3 B

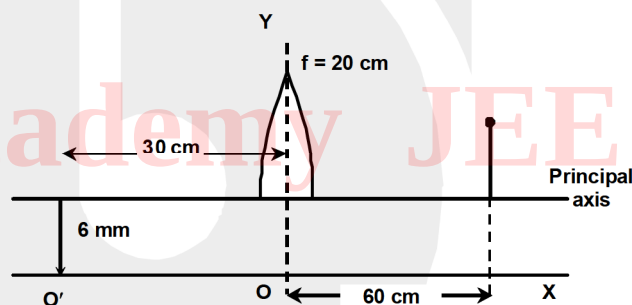
Sol. $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\Rightarrow \frac{1}{v} = \frac{1}{u} + \frac{1}{f} = -\frac{1}{30} + \frac{1}{20} = \frac{3-2}{60} = \frac{1}{60}$$

$$\Rightarrow v = 60 \text{ cm}$$

$$m = v/u = 60/-30 = -2$$

Height of image = $(-2)(-6) = 12 \text{ mm}$
 Y-coordinate of the image of $Q_1 = 12 + 6$
 $= 18 \text{ mm}$



Q.4 C

Sol. $P = 2P_l + P_m$

$$P_m = \frac{1}{f}, \frac{1}{f_l} = \left(\frac{\mu_g}{\mu_l} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right). \text{ Converging mirror.}$$

Q.5 A

Sol. $v = \frac{c}{kx}$

$$v = \frac{c}{kx}$$

$$\int_0^t x \, dx = \frac{c}{k} \int_0^T dt$$

$$T = \frac{t^2 k}{2c}$$

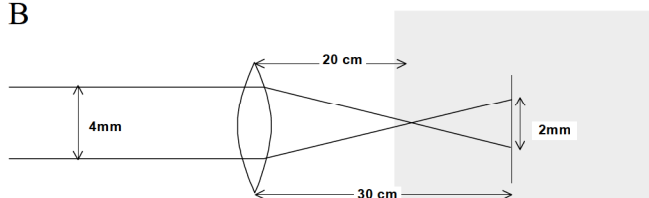
Q.6 B

Sol. Only 'z' component will change.

Q.7 D

Q.8 B

Sol.



Q.9 B

Q.10 D

Q.11 B

Q.12 D

Q.13 C

Q.14 A

Q.15 D

Q.16 B

Sol. Mirror formula

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

$$-\frac{1}{u^2} \frac{du}{dt} - \frac{1}{v^2} \frac{dv}{dt} = 0$$

$$\left| \frac{dv}{dt} \right| = \frac{v^2}{u^2} \left| \frac{du}{dt} \right| = \left(\frac{10}{20} \right)^2 \times 4 = 1 \text{ m/sec}$$

Q.17 C

Sol. Net dispersion produced = $(\delta_v - \delta_r) - (\delta_{v'} - \delta_{r'}) = (\mu_v - \mu_r)A - (\mu_{v'} - \mu_{r'})B$

Q.18 A

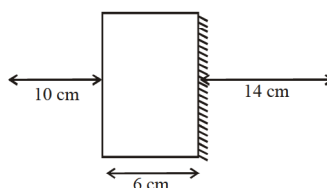
Sol. Sol. For diverging lens, all refracted rays corresponding to an incident beam consisting of parallel rays meet at a point in focal plane when produced backward

Q.19 A

Sol. Dist. between object and image

$$30 = 2 \left(10 + \frac{6}{\mu} \right)$$

$$\Rightarrow \mu = 1.2$$

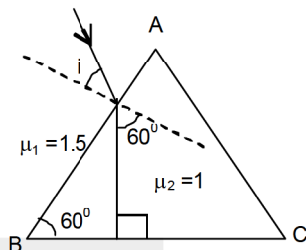


Q.20 A

Sol. $1.5 \sin i = (1) \sin 60^\circ$

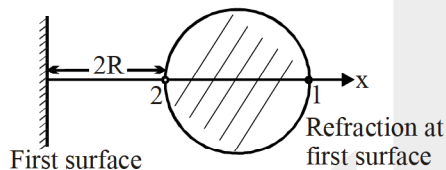
$$\Rightarrow \frac{3}{2} \cdot \sin i = \frac{\sqrt{3}}{2}$$

$$\therefore i = \sin^{-1} \left(\frac{1}{\sqrt{3}} \right)$$



Q.21 [Ans. (A) P,R (B) Q,R (C) Q,S]

[Sol.



$$\frac{\mu_r}{v} - \frac{\mu_i}{u} = \frac{\mu_r - \mu_i}{R}$$

$$\frac{1}{v} - \frac{4/3}{(-2R)} = \frac{1 - 4/3}{(-R)}$$

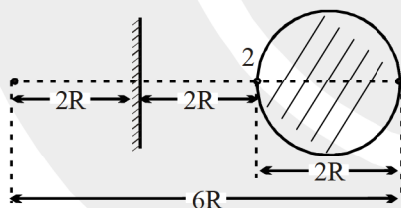
$$\frac{1}{v} + \frac{2}{3R} = -\frac{1}{3R}$$

$$v = -3R \quad (\text{virtual})$$

$$m = \frac{\mu_i}{\mu_r} = \frac{4/3}{(1)} \frac{(-3R)}{(-2R)}$$

$$m = 2 \text{ (magnified)}$$

Reflection from mirror & refraction from second surface

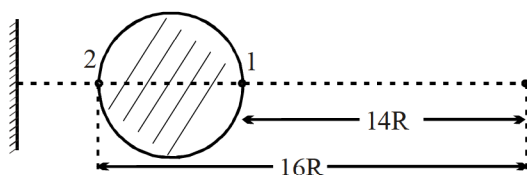


$$\frac{4/3}{v} - \frac{1}{(-4R)} = \frac{4/3 - 1}{R}$$

$$v = 16R \quad (\text{Real})$$

$$m = \frac{\mu_i}{\mu_r} \frac{v}{u} = \frac{1}{4/3} \frac{16R}{(-4R)} \Rightarrow m = -3 \text{ (magnified)}$$

Refraction a first surface after reflection from mirror and refraction at second surface.



$$\frac{\mu_r}{v} - \frac{\mu_i}{u} = \frac{\mu_r - \mu_i}{R}$$

$$\frac{1}{v} - \frac{4/3}{14R} = \frac{1 - 4/3}{-R}$$

$$v = \frac{7R}{3} \text{ (Real)}$$

$$m = \frac{\mu_i}{\mu_r} \frac{v}{u}$$

$$m = \frac{4}{3} \times \frac{7/3R}{+14R} = \frac{4}{3} \times \frac{7}{3 \times 14} \Rightarrow m = \frac{2}{9} \text{ (diminished)]}$$

Q.22 [Ans. $\mu = \frac{1}{2 \sin 15^\circ}$]

Sol. Also $R = 1\text{m}$
In triangle PQO

$$\frac{PQ}{PO} = \sin 2r$$

$$\frac{0.5}{1} = \sin 2r$$

$$\Rightarrow 2r = 30^\circ \text{ or } r = 15^\circ$$

It is clear from figure that

$$i = 2r = 30^\circ$$

$$\text{so } \mu = \frac{\sin i}{\sin r} = \frac{\sin 30}{\sin 15} = \frac{1}{2 \sin 15^\circ}$$

Q.23 [Ans. $2r\omega \sin 2\theta$]

Sol. Velocity of point Q w.r.t. mirror is

$$= (r\omega + r\omega \cos 2\theta)\hat{i} + (r\omega \sin 2\theta)(-\hat{j})$$

Horizontal velocity of image is same but vertical velocity is in opposite direction

so velocity of image w.r.t. Q in vertical direction is $= 2r\omega \sin 2\theta = 4r\omega \sin \theta \cos \theta$

